

UML DIAGRAMS FOLDER

Overview

This UML Diagrams folder contains all graphical models of the Smart Parking System. These diagrams represent the system’s structure, behavior, and interactions in a standardized format. They help in understanding the system design before implementation and ensure consistency across all components.

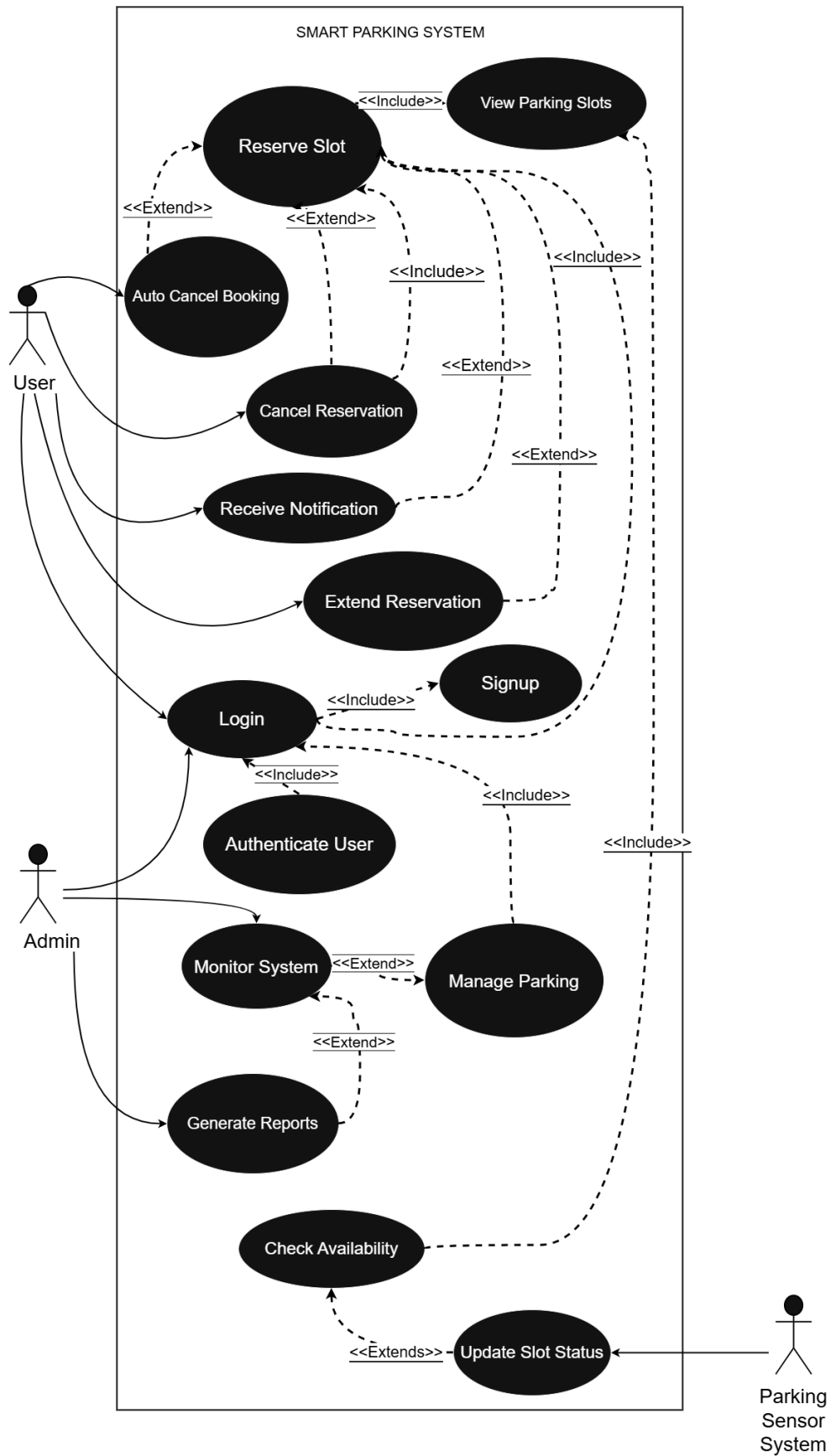
Included UML Diagrams

Diagram Type	Purpose	Description
Use Case Diagram	Functional View	Shows interactions between users and system
Class Diagram	Structural View	Defines system classes and relationships
Sequence Diagram	Interaction Flow	Shows step-by-step process of reservation
Activity Diagram	Workflow	Represents system processes and decisions
State Machine Diagram	State Behavior	Shows lifecycle of parking slots

Key Components

Use Case Diagram

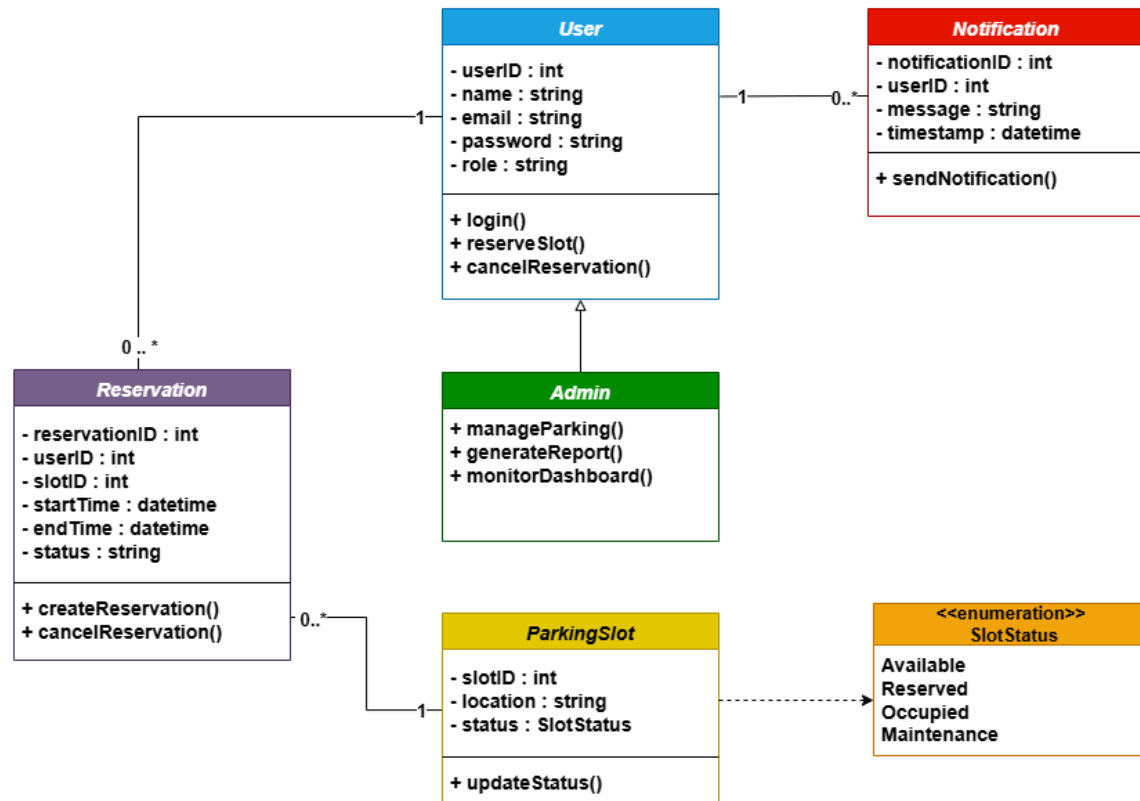
- Identifies system actors (Student, Admin)
- Defines system functionalities such as booking and slot checking



Class Diagram

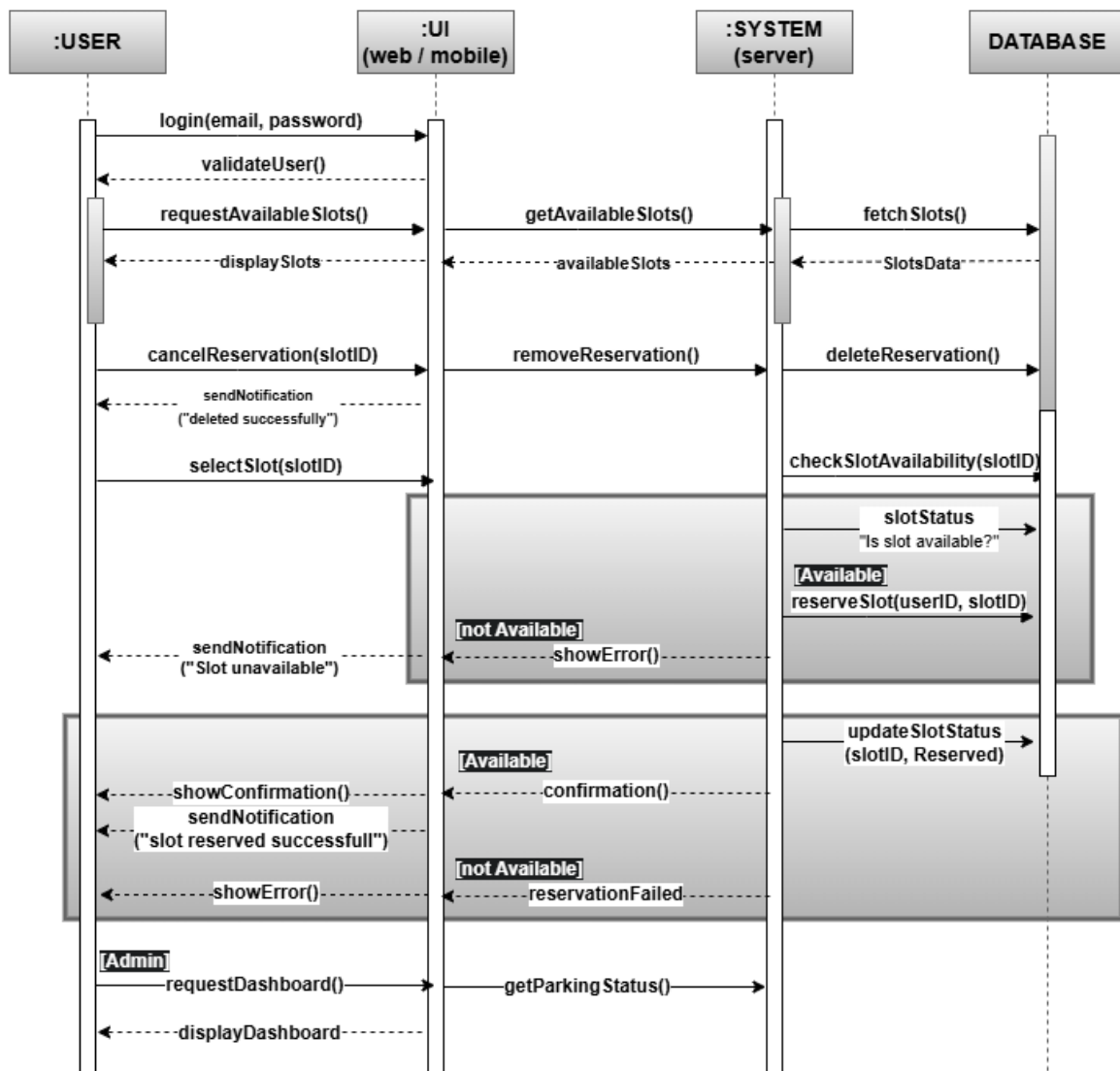
- Represents system structure
- Includes classes such as User, Admin, ParkingSlot, Reservation

SMART PARKING SYSTEM FOR UNIVERSITY CAMPUS
UML CLASS DIAGRAM



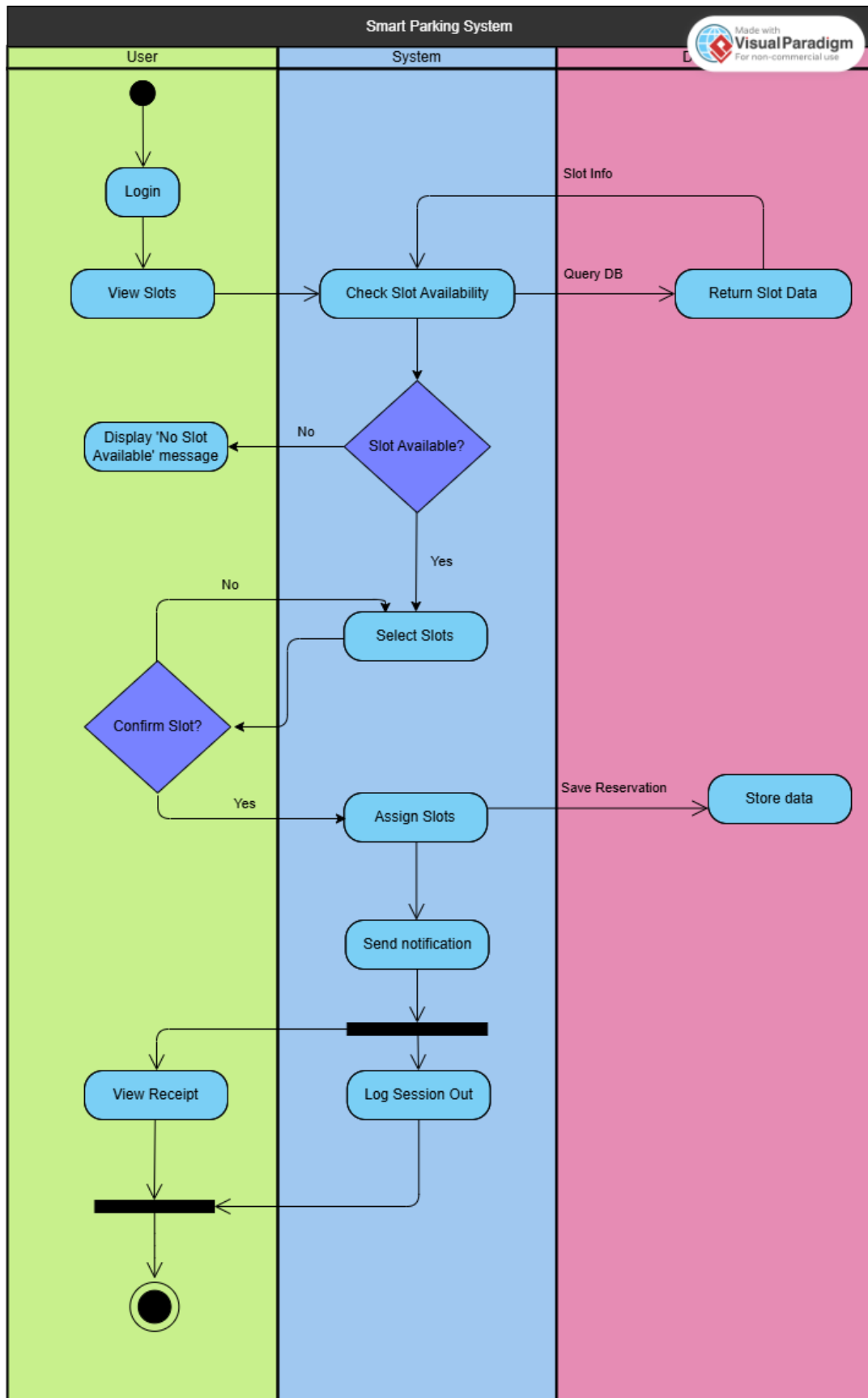
Sequence Diagram

- Shows interaction flow between system components during booking process



Activity Diagram

- Represents workflow of system operations step by step



State Machine Diagram

- Shows states of parking slots such as Available, Reserved, and Occupied

